BERTScore: Evaluating Text Generation with BERT

Zhang, Kishore, Wu, Weinberger and Artzi (ICLR 2020), arXiv:1904.09675

Paper summary for the CS-781 capstone

The paper at a glance

Citation

T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, Y. Artzi. *BERTScore: Evaluating Text Generation with BERT*. ICLR 2020. Cornell University and ASAPP.

One-sentence summary

Score generated text by matching each token to its most similar token in the reference, using cosine similarity of *contextual embeddings* instead of exact word overlap.

Why it matters for our capstone

BERTScore F1 is the **primary metric** for Phase 1 (continuations) and Phase 6 (summaries). Our `06_evaluate.py` computes it locally.

Where it appears in our slides

Main deck: "Step 06: the metrics". Glossary: "BERTScore" slide.

Roadmap

1. The problem
2. The basic idea
3. Experiments
4. Assessment and outlook

What is wrong with n-gram metrics?

BLEU, ROUGE and METEOR count overlapping words or n-grams between a candidate and a reference. Two failures:

1. **Paraphrases are punished.** Correct meaning in different words scores low.
2. **Order and long-range meaning are missed.** With short n-grams, swapping cause and effect ("A because B" vs "B because A") is only mildly penalised.

Learned metrics (e.g. RUSE, BEER) fix some of this, but need human judgement data for training and may not transfer to new domains.

The paper's example

Reference: *people like foreign cars.*

- A: *people like visiting places abroad*
- B: *consumers prefer imported cars*

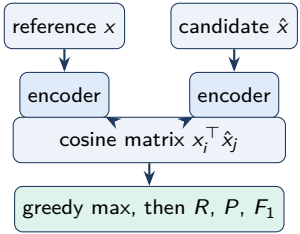
BLEU and METEOR score A **higher** than B, although B means the same thing and A does not.

A legal version

Reference: *the appeal is dismissed.* Candidate: *the petition is rejected.* Word overlap sees only "the" and "is".

BERTScore in three steps

1. **Embed**: run reference x and candidate $\hat{x}$ through a pretrained encoder (BERT, RoBERTa, ...). Each token gets a *contextual* vector that depends on its sentence.
2. **Compare**: compute the cosine similarity of every reference token with every candidate token. Vectors are pre-normalised, so this is a dot product.
3. **Match greedily**: each token takes its best match on the other side. Average these to get recall and precision, then F1.



The formulas

$$R_{\text{BERT}} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} x_i^\top \hat{x}_j$$

$$P_{\text{BERT}} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} x_i^\top \hat{x}_j$$

$$F_{\text{BERT}} = 2 \frac{P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}}$$

- **Recall**: is every reference token covered by something similar in the candidate?
- **Precision**: is every candidate token supported by something in the reference?

Two optional extras

idf weighting: rare words matter more.

$$\text{idf}(w) = -\log \tfrac{1}{M} \textstyle\sum_{i=1}^{M} \mathbb{I}[w \in x^{(i)}]$$

computed on the M test references, with plus-one smoothing.

Baseline rescaling: raw scores cluster in a narrow range. Subtract the average score b of random sentence pairs:

$$\hat{R} = \frac{R - b}{1 - b}$$

This only improves readability; rankings are unchanged.

Easy worked example

Reference: *appeal dismissed with costs*
Candidate: *petition rejected with costs*

Illustrative cosine similarities (content words only):

	petition	rejected	costs
appeal	**0.80**	0.30	0.20
dismissed	0.30	**0.85**	0.25
costs	0.20	0.25	**1.00**

Numbers chosen for illustration, not computed by a model.

Scoring

Recall: row maxima $0.80, 0.85, 1.00$, so $R = 0.883$.
Precision: column maxima $0.80, 0.85, 1.00$, so $P = 0.883$.
$F_1 = 0.883$.

Exact unigram matching finds only "costs": $P = R = 1/3$. BERTScore rewards the right *meaning*.

Watch out

"Allowed" and "dismissed" occur in similar contexts, so a reversed outcome may lose less score than it should.

Design choices and the paper's recommendations

- **12 embedding models** were compared: BERT, RoBERTa, XLNet and XLM variants.
- **Which layer?** Intermediate layers work better than the last one. The best layer per model was chosen on WMT16.
- **Greedy vs optimal matching**: greedy is simpler and performed comparably (appendix).
- **Tokenisation**: each model's own word pieces.

Recommendations

- Use **F1**: recall, precision and F1 alternate as best, but F1 is reliably good.
- English: **24-layer RoBERTa-large**.
- Other languages: multilingual BERT, which is less stable for low-resource languages.
- idf weighting sometimes helps slightly and sometimes not.

Our local setup

`bert_score` with `lang="en"` defaults to RoBERTa-large, matching this advice.

How a metric is evaluated

A good metric **agrees with human judgements**. Tests used:

- **System level**: Pearson correlation between each system's average metric score and its human score.
- **Segment level**: Kendall correlation per sentence. Much harder.
- **Model selection**: in 100 random systems, does the metric pick the same best system as humans (Hits@1)?
- **Robustness**: can it tell real paraphrases from adversarial word swaps?

Data

- WMT18 translation metrics task: 149 systems, 14 language directions. WMT16–17 add 202 more.
- COCO 2015 image captioning: 12 systems.
- QQP and PAWS paraphrase pairs.

363 systems in total across MT and captioning.

Machine translation: segment-level agreement

Kendall τ with human judgements on WMT18 (to-English / from-English):

Pair	BLEU	YiSi-1	F_{BERT}
cs	.233/.389	.319/.496	**.404/.562**
de	.415/.620	.488/.691	**.550/.728**
et	.285/.414	.351/.546	**.397/.586**
fi	.154/.355	.231/.504	**.296/.546**
ru	.228/.330	.300/.407	**.353/.423**
tr	.145/.261	.234/.418	**.292**/.399
zh	.178/.311	.211/.323	**.264/.364**

Reading it

- BERTScore beats BLEU in every direction at the sentence level, often by a wide margin.
- It even beat RUSE, a metric *trained* on human judgements.
- At the *system* level most metrics already correlate at 0.95 or more, so gains there are smaller.

So BERTScore is most useful for judging **individual outputs**.

Image captioning and paraphrase robustness

COCO 2015, Pearson correlation with two human measures (M1 / M2):

Metric	M1	M2
BLEU	−0.019	−0.005
ROUGE-L	0.090	0.096
CIDEr	0.438	0.440
SPICE (task-specific)	0.759	0.750
R_{BERT}	0.888	0.863
R_{BERT} + idf	**0.917**	**0.889**

Only LEIC, which is trained on COCO and sees the images, scored higher.

Paraphrase detection, AUC (QQP / adversarial PAWS):

Method	QQP	PAWS
BERT trained on QQP	0.963	0.351
BLEU	0.707	0.527
METEOR	0.755	0.532
F_{BERT}	0.761	0.685
F_{BERT} + idf	**0.777**	**0.693**

PAWS swaps words: “flights from New York to Florida” vs “from Florida to New York”. n-gram metrics fall to near chance (0.5); BERTScore drops much less.

Speed

Despite running a large model, BERTScore is practical:

- About **192.5 candidate–reference pairs per second** on one GTX-1080Ti GPU.
- The WMT18 EN–DE test set (2,998 sentences) takes **15.6 s**, against 5.4 s for SacreBLEU.

The authors conclude it is cheap enough to use during validation, e.g. for early stopping, as well as for testing.

For our dev set

200 dev continuations of 128 words each are a few hundred pairs. Scoring them takes seconds on a GPU. The one-time download of RoBERTa-large (about 1.4 GB) costs more than the scoring itself.

What this means for the capstone

- Phase 1 and Phase 6 are ranked by BERTScore F1, so a continuation that **paraphrases** the hidden text well can score high even with few exact word matches.
- Content words, such as parties, statutes and outcomes, count more than style.
- Decoding choices matter: very random sampling adds off-topic tokens that lower precision; very repetitive text lowers recall.

Experiments to run

- Compare dev BERTScore with dev loss: do they rank checkpoints the same way?
- Temperature and top-p sweeps, scored by BERTScore.
- Report P, R and F_1 separately to see *why* a model scores as it does.
- Check a few outputs with reversed outcomes to see how much BERTScore penalises them.

Strengths and limitations

Strengths

- Rewards paraphrases and meaning, not surface form.
- Task-agnostic; needs no human-judgement training data.
- Much better sentence-level agreement with humans than BLEU.
- More robust to adversarial word swaps.
- Available in 104 languages via multilingual BERT.
- Fast enough for routine use.

Limitations

- Scores depend on the model, layer and version, so they are not comparable across setups.
- No single configuration wins everywhere.
- Greedy token matching ignores global structure.
- Unstable for some low-resource settings: recall for Turkish→English correlated only 0.054 at system level.
- Measures similarity, not **factual correctness**: a fluent wrong outcome may still score well.

Future research directions (1): from the paper and after

Direction	Details
Task-specific metrics built on BERTScore *(paper)*	Use BERTScore as a subroutine, adapted to each task's needs
BERTScore as a training loss *(paper)*	It is differentiable, so it can reduce the gap between the training objective and evaluation
When does idf weighting help? *(paper)*	Likely depends on domain and test data; left open
Optimal-transport matching	MoverScore (Zhao et al., 2019) uses earth mover's distance over contextual embeddings
Learned metrics on pretrained models	BLEURT (Sellam et al., 2020) and COMET (Rei et al., 2020) are fine-tuned on human ratings
Generation-based and LLM judges	BARTScore (Yuan et al., 2021); LLM-as-judge methods such as G-Eval (Liu et al., 2023)
Domain-specific embeddings	The paper notes SciBERT-based BERTScore for scientific summaries (Gabriel et al., 2019)

Future research directions (2): open questions

Open in general

- Detecting **factual errors and negation** with embedding metrics.
- Evaluating long outputs: whole summaries, not sentences.
- Calibrated, comparable scores across models and versions.
- Metrics that resist being gamed when used as training targets.

Open for legal NLP

- Would a **legal-domain encoder** give BERTScores that agree better with lawyers' judgements?
- Weighting legally critical tokens: outcome, statute, party, date.
- Pairing BERTScore with an outcome-consistency check for summaries and continuations.
- Building a small human-rated legal set to measure how well BERTScore agrees with experts.

Summary

- n-gram metrics punish paraphrases and miss long-range meaning.
- **BERTScore**: contextual embeddings, a cosine-similarity matrix, greedy max matching, then R, P and F_1. Optional idf weighting and baseline rescaling.
- It agrees with humans much better than BLEU at the sentence level (e.g. EN–DE τ .550 vs .415), and ranks near the top at the system level.
- Strong on captioning (0.917 vs SPICE 0.759) and robust to adversarial paraphrases.
- Recommendation: F1 with RoBERTa-large for English.
- For us: the metric that ranks Phase 1 and Phase 6, so tune decoding against it on the dev set.

Reference

Zhang, T., Kishore, V., Wu, F., Weinberger, K. Q., and Artzi, Y. (2020). BERTScore: Evaluating Text Generation with BERT. In *International Conference on Learning Representations (ICLR)*. arXiv:1904.09675.

Other work cited: Papineni et al. (2002) BLEU; Lin (2004) ROUGE; Banerjee and Lavie (2005) METEOR; Zhao et al. (2019) MoverScore; Gabriel et al. (2019); Sellam et al. (2020) BLEURT; Rei et al. (2020) COMET; Yuan et al. (2021) BARTScore; Liu et al. (2023) G-Eval.